

THE MIRROR SEDENIONS: A SECOND G_2 -SYMMETRIC DOUBLING OF THE OCTONIONS AND THE GEOMETRY OF ITS ZERO DIVISORS

LUI

ABSTRACT. The Cayley–Dickson double of the octonions may be formed with the product of the founding copy of $\mathbb{O}$ written backwards:

$$(a, b)(c, d) = (ca - \bar{d}b, da + b\bar{c}).$$

For a commutative base this changes nothing; over $\mathbb{H}$ it already produces the quasi-octonions instead of $\mathbb{O}$; over $\mathbb{O}$ it produces a 16-dimensional real algebra $\mathbb{S}'$, the *mirror sedenions*, which is flexible, quadratic, carries the diagonal action of G_2 , and is not isomorphic to the sedenions $\mathbb{S}$. We prove that $\mathbb{S}'$ is the subalgebra $\mathbb{O} + \mathbb{O}(ee')$ of the 32-dimensional Cayley–Dickson algebra spanned by $\mathbb{O}$ and the composite unit ee' (the algebra noticed but not studied by Cawagas et al.), that Bales’s eight admissible doubling products applied to $\mathbb{O}$ yield exactly the two algebras $\mathbb{S}$ and $\mathbb{S}'$, and we determine the structure of $\mathbb{S}'$ completely. The stretch operator $L_{\bar{x}}L_x$ has the five-level spectrum $N(x) \pm 2|\operatorname{Im} a||b|$, $N(x) \pm 2|\operatorname{Im} a||b_{\parallel}|$, $N(x)$ with multiplicities 2, 2, 4, 4, 4, where $x = a + b\ell$ and $b_{\parallel}$ is the projection of b to the complex line $\mathbb{R} + \mathbb{R}\operatorname{Im} a$. Consequently the norm-one zero divisors of $\mathbb{S}'$ form the product of round spheres $S^6 \times S^7$ (a 13-manifold, against the Stiefel manifold $V_2(\mathbb{R}^7)$ of Moreno and Biss–Dugger–Isaksen for $\mathbb{S}$), annihilators are 2-dimensional except on a 7-dimensional locus where they are 6-dimensional, and the manifold of norm-one zero-divisor pairs is diffeomorphic to $V_2(\mathbb{R}^7) \times S^3$, the *trivial* S^3 -bundle over the Stiefel manifold, whereas for $\mathbb{S}$ it is G_2 , the principal $SU(2)$ -bundle over the same base. The eight octonion subalgebras spanned by basis hyperplanes are the eight hyperplanes of $PG(3, 2)$ avoiding the doubling point, whereas for $\mathbb{S}$ they are the founding octave and the seven hyperplanes through the doubling point; the seven remaining hyperplanes are quasi-octonion algebras in both cases. $\operatorname{Aut}(\mathbb{S}') = G_2 \times \mathbb{Z}/2$, in contrast with Brown’s $G_2 \times S_3$ for $\mathbb{S}$. The difference between the two algebras is a single sign function χ on $\mathbb{F}_2^4 \times \mathbb{F}_2^4$ which is not a 2-cocycle: the “orientation bit” of the fourth doubling.

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1. INTRODUCTION

Let $\mathbb{O}$ be the real octonions with conjugation $x \mapsto \bar{x}$ and norm $N(x) = x\bar{x}$. The Cayley–Dickson double of $\mathbb{O}$ is the sedenion algebra $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}\ell$ with

$$(1) \quad (a + b\ell)(c + d\ell) = (ac - \bar{d}b) + (da + b\bar{c})\ell.$$

The literature contains many conventions for this formula. Bales [2] lists eight “doubling products” that yield a Cayley–Dickson algebra when applied at every level starting from $\mathbb{R}$, and notes that the four “transposed” products give the opposite algebras of the other four. The present paper begins with a different question: what happens when one of these eight products is applied *once*, on top of the standard octonions? The answer is that exactly two algebras arise. Four of Bales’s products give $\mathbb{S}$. The other four give the algebra

$$(2) \quad (a + b\ell)(c + d\ell) = (ca - \bar{d}b) + (da + b\bar{c})\ell,$$

which differs from (1) in a single slot: the product of the two octonions from the founding copy of $\mathbb{O}$ is taken backwards. We call this algebra the *mirror sedenions* and denote it $\mathbb{S}'$. For any base algebra A we write $M(A)$ for the algebra $A \oplus A$ with product (2) and call it the *mirror double* of A ; thus $\mathbb{S}' = M(\mathbb{O})$.

If A is commutative, $M(A) = CD(A)$. If A is associative but not commutative the two doubles are already different: $M(\mathbb{H})$ is not a composition algebra. Over the octonions the change of a single letter produces a genuinely new object with a rich and very clean geometry, which is the subject of this paper. Our main results, all proved by hand and independently verified by computer, are the following. Throughout, $x = a + b\ell \in \mathbb{S}'$ with $a, b \in \mathbb{O}$, $N(x) = |a|^2 + |b|^2$, and $b_{\parallel}$ denotes the orthogonal projection of b onto the complex line $\mathbb{C}_a = \mathbb{R} + \mathbb{R} \operatorname{Im} a$.

- (A) *Embedding* (Theorem 3.3). For every unital $*$ -algebra A , the mirror double $M(A)$ is isomorphic to the subalgebra $A + A(ee')$ of the double double $CD(CD(A))$ spanned by A and the product ee' of the two doubling units, via $a + b\ell \mapsto \bar{a} + (be)e'$. Hence $M(\mathbb{H})$ is the quasi-octonion algebra $\mathbb{H} + \mathbb{H}(\varepsilon\ell) \subset \mathbb{S}$ ($\varepsilon \perp \mathbb{H}$ a unit octonion) of Cawagas [8] and Chan–Đoković [10], and $\mathbb{S}'$ is the 16-dimensional subalgebra $\mathbb{S}^{\gamma} = \mathbb{O} + \mathbb{O}(ee')$ of the trigintaduonions which Cawagas et al. [9] found to be the unique basis hyperplane of $\mathbb{T}$ with eight octonion subalgebras not isomorphic to $\mathbb{S}$.
- (B) *The two doublings* (Theorem 3.6). Applied to $\mathbb{O}$, Bales’s eight products give $\mathbb{S}$ (for P_0, P_3, P_0^T, P_3^T) or $\mathbb{S}'$ (for P_1, P_2, P_1^T, P_2^T), with explicit isomorphisms of the form $a + b\ell \mapsto a^{(\cdot)} + (\pm b^{(\cdot)})\ell$. The two classes are exchanged by reversing the first product $ac \leftrightarrow ca$.
- (C) *Spectral theorem* (Theorem 4.1). The eigenvalues of $L_{\bar{x}}L_x$ on $\mathbb{S}'$ are

$$N(x) \pm 2|\operatorname{Im} a||b| \text{ (multiplicity 2 each),} \quad N(x) \pm 2|\operatorname{Im} a||b_{\parallel}| \text{ (4 each),} \quad N(x) \text{ (4).}$$

For $\mathbb{S}$ the analogous spectrum [5] has three levels $N \pm 2|\operatorname{Im} a \times \operatorname{Im} b|$, N with multiplicities 4, 4, 8.

- (D) *Zero divisors* (Theorems 5.1, 5.2). x is a zero divisor if and only if $\operatorname{Re} a = 0$ and $|a| = |b|$. The norm-one zero divisors form the Riemannian product $S^6(\frac{1}{\sqrt{2}}) \times S^7(\frac{1}{\sqrt{2}})$, a 13-manifold. For $u \in S^6 \subset \operatorname{Im} \mathbb{O}$ and $b \in S^7 \subset \mathbb{O}$,

$$\operatorname{Ann}(u + b\ell) = \{ nu + (bn)\ell : n \in \operatorname{Im} \mathbb{O}, n \perp u, [b, n, u] = 0 \},$$

which has dimension 2 unless $b \in \mathbb{C}_u$, when it has dimension 6.

- (E) *The pair manifold* (Theorem 5.5). The set $\mathcal{P}(\mathbb{S}')$ of pairs (x, y) of norm-one elements with $xy = 0$ is a smooth closed 14-manifold, G_2 -equivariantly diffeomorphic to $V_2(\mathbb{R}^7) \times S^3$ with G_2 acting on the first factor; every stabilizer is $SU(2)$ and the orbit space is S^3 . It fibres over the Grassmannian $G_2/SO(4)$ of quaternion subalgebras with fibre $SO(3) \times S^3$. It is not homotopy equivalent to Moreno's $\mathcal{P}(\mathbb{S}) \cong G_2$.
- (F) *Octaves* (Theorem 6.1). Of the fifteen 8-dimensional subalgebras spanned by basis hyperplanes of $\mathbb{F}_2^4$, the eight that are octonion algebras are exactly the hyperplanes not containing the doubling point ℓ ; for $\mathbb{S}$ they are $\mathbb{O}$ together with the seven hyperplanes through ℓ . The seven others are quasi-octonion algebras $M(\mathbb{H})$ in both cases.
- (G) *Symmetry* (Theorem 7.1). $\operatorname{Aut}(\mathbb{S}') = G_2 \times \mathbb{Z}/2$ and $\operatorname{Der}(\mathbb{S}') = \mathfrak{g}_2$, acting diagonally; the order-three automorphism of $\mathbb{S}$ found by Brown [7] does not survive. The alternative elements of $\mathbb{S}'$ are $\mathbb{O} \cup (\mathbb{R} + \mathbb{O}\ell)$.
- (H) *The orientation bit* (Propositions 7.3, 7.4). As twisted group algebras of $\mathbb{F}_2^4$, the sign functions of $\mathbb{S}$ and $\mathbb{S}'$ differ by the function χ which is -1 on pairs of distinct nonzero points of the founding $\mathbb{F}_2^3$ and $+1$ elsewhere; χ is not a 2-cocycle, and no signed relabelling of the sixteen basis units carries $\mathbb{S}$ to $\mathbb{S}'$.

The picture that emerges is a precise duality. Both algebras contain eight octonion subalgebras and seven quasi-octonion subalgebras among their basis hyperplanes, both have derivation algebra $\mathfrak{g}_2$, both have a 14-dimensional manifold of zero-divisor pairs fibring over $V_2(\mathbb{R}^7)$ with fibre S^3 ; but in $\mathbb{S}$ the frame $(u, v) \in V_2(\mathbb{R}^7)$ is carried by the zero divisor $x = u + v\ell$ itself and the S^3 by its annihilator, so that $\mathcal{P}(\mathbb{S})$ is the principal bundle $G_2 \rightarrow G_2/SU(2)$, while in $\mathbb{S}'$ the frame (u, n) is split between $x = u + b\ell$ and its partner $y = nu + (bn)\ell$, the S^3 is the quaternion coordinate of b in the chart $\mathbb{H}(u, n)$, and the bundle is trivial. Moreno's theorem that the zero-divisor pairs of $\mathbb{S}$ form a G_2 -torsor [18] is, in this light, a statement about the alignment of the orientation of the founding octave with the doubling twist, not about G_2 alone: $\mathbb{S}'$ has the same G_2 and no torsor.

Relation to the literature. The formula (2) is Bales's product P_1^T [2]; Bales considers only towers in which one product is used at every level, and the uniform P_1^T tower is a different algebra (its dimension-8 stage is already $M(\mathbb{H})$ rather than $\mathbb{O}$, and its dimension-16 stage has a 6-dimensional derivation algebra; Remark 3.8). The algebra $\mathbb{S}'$ itself appears, as the subalgebra $\mathbb{S}^\gamma \subset \mathbb{T}$, in the loop-theoretic census of Cawagas et al. [9], who record that it has eight octonion and seven quasi-octonion subloops and is not isomorphic to $\mathbb{S}$, and prove nothing further. The zero-divisor theory of $\mathbb{S}$ is due to Moreno [18], Biss–Dugger–Isaksen [3, 4] and Biss–Christensen–Dugger–Isaksen [5]; its Riemannian geometry to Reggiani [19]; the automorphism group to Brown [7] (see also Eakin–Sathaye [13]); the derivation algebra to Schafer [22]; the finite geometry of its multiplication table to Saniga–Holweck–Pracna [21], Cawagas [8] and de Marrais [11]; the twisted-group-algebra viewpoint to Albuquerque–Majid [1]. We found no prior description of the zero divisors, annihilators, spectrum, automorphisms, or subalgebra geometry of $\mathbb{S}'$, and no prior formulation of the mirror double as a construction.

Verification. Every numerical statement in this paper (dimensions of kernels and tangent spaces, the census of the thirty-two products, the graded isomorphism search over $GL(4, 2)$, the symbolic characteristic polynomials) was checked with the scripts listed in Appendix A. The proofs themselves do not depend on the computer.

2. CONVENTIONS AND OCTONION FACTS

2.1. Cayley–Dickson algebras. A $*$ -algebra is a unital real algebra A with a linear involution $x \mapsto \bar{x}$ satisfying $\overline{\bar{y}} = y$ and $\bar{1} = 1$. Its *Cayley–Dickson double* is $CD(A) = A \oplus Ae$ with the product (1) (with ℓ replaced by e) and involution $\overline{a + be} = \bar{a} - be$. Starting from $\mathbb{R}$ with the trivial involution one obtains $\mathbb{C}, \mathbb{H}, \mathbb{O}, \mathbb{S}, \mathbb{T}, \dots$; we write $\mathbb{O} = CD(\mathbb{H})$ with doubling unit e_4 , $\mathbb{S} = CD(\mathbb{O})$ with doubling unit $\ell = e_8$, and $\mathbb{T} = CD(\mathbb{S})$ with doubling unit $e' = e_{16}$.

Convention 2.1. The basis of $\mathbb{O}$ is $e_0 = 1, e_1, \dots, e_7$ with $e_g e_h = \pm e_{g \oplus h}$ ($\oplus$ = bitwise XOR), e_1 the imaginary unit of $\mathbb{C}$, e_2 the doubling unit of $\mathbb{H}$, e_4 that of $\mathbb{O}$. With (1) the oriented lines of the Fano plane are

$$e_1 e_2 = e_3, \quad e_1 e_4 = e_5, \quad e_2 e_4 = e_6, \quad e_3 e_4 = e_7, \quad e_1 e_7 = e_6, \quad e_2 e_5 = e_7, \quad e_3 e_6 = e_5.$$

In $\mathbb{S}$ and $\mathbb{S}'$ we write $e_{g+8} = e_g \ell$ for $0 \leq g \leq 7$, so that both algebras have basis $e_0, \dots, e_{15}$ indexed by $\mathbb{F}_2^4$, and $e_g e_h = \pm e_{g \oplus h}$ in both. Note that inside $\mathbb{S}'$ the founding copy of $\mathbb{O}$ carries the opposite product: $e_1 e_2 = -e_3$ in $\mathbb{S}'$. We identify the subspace $\mathbb{O} = \{a + 0\ell\}$ with the octonions via $a \mapsto \bar{a}$ when an algebra isomorphism is needed, and otherwise simply regard a, b in $x = a + b\ell$ as octonions multiplied in $\mathbb{O}$.

For $x = a + b\ell$ in $\mathbb{S}$ or $\mathbb{S}'$ put $\bar{x} = \bar{a} - b\ell$, $N(x) = x\bar{x} = |a|^2 + |b|^2$ and $\langle x, y \rangle = \frac{1}{2}(N(x + y) - N(x) - N(y))$, the standard inner product of $\mathbb{R}^{16}$. We write L_x, R_x for left and right multiplication, $[x, y, z] = (xy)z - x(yz)$ for the associator, and $\text{alt}_x = [x, x, \cdot]$ for the *alternator* of x .

2.2. Octonion identities. We use freely: $\mathbb{O}$ is alternative, so any two elements generate an associative subalgebra (Artin) which is contained in a quaternion subalgebra; the Moufang identities; $\overline{xy} = \bar{y}\bar{x}$; $x\bar{x} = |x|^2$; the adjoint identity $\langle xy, z \rangle = \langle y, \bar{x}z \rangle = \langle x, z\bar{y} \rangle$; the associator $[x, y, z]$ is alternating and orthogonal to $1, x, y, z$; and $(xy)\bar{y} = |y|^2 x$, $\bar{x}(xy) = |x|^2 y$. For $x \in \mathbb{O}$ we write $x = x_0 + x'$ with $x_0 = \text{Re } x \in \mathbb{R}$ and $x' = \text{Im } x \in \text{Im } \mathbb{O}$, and $\mathbb{C}_x = \mathbb{R} + \mathbb{R}x'$ for the complex line through x . For $u \in \text{Im } \mathbb{O}$ a unit, an element y satisfies $yu = -uy$ if and only if $y \in \text{Im } \mathbb{O}$ and $y \perp u$.

Lemma 2.2. *Let $u, b \in \mathbb{O}$ with $b \notin \mathbb{C}_u$ and $u \notin \mathbb{R}$. Then $\{n \in \mathbb{O} : [b, n, u] = 0\}$ is the quaternion subalgebra $Q = \mathbb{H}(u, b)$ generated by u and b .*

Proof. $Q \subseteq \ker$ by Artin. Write $\mathbb{O} = Q \oplus Q\varepsilon$ with $\varepsilon \perp Q$ a unit, so that $\mathbb{O} = CD(Q)$ with the product (1). For $q \in Q$ one computes from (1): $b(q\varepsilon) = (qb)\varepsilon$, $(q\varepsilon)u = (q\bar{u})\varepsilon$, hence

$$[b, q\varepsilon, u] = ((qb)\bar{u} - (q\bar{u})b)\varepsilon = q(b\bar{u} - \bar{u}b)\varepsilon.$$

Since $b\bar{u} - \bar{u}b = -(bu - ub) = -2b' \times u'$ is a nonzero quaternion when $b \notin \mathbb{C}_u$, this vanishes only for $q = 0$. As the map $n \mapsto [b, n, u]$ is linear and $\mathbb{O} = Q \oplus Q\varepsilon$, the kernel is Q . $\square$

2.3. Facts about G_2 . $G_2 = \text{Aut}(\mathbb{O})$ acts on $\text{Im } \mathbb{O} \cong \mathbb{R}^7$ preserving the inner product and the cross product. It acts transitively on the unit sphere S^6 with stabilizer $SU(3)$, transitively on the Stiefel manifold $V_2(\mathbb{R}^7)$ of orthonormal pairs in $\text{Im } \mathbb{O}$ with stabilizer $SU(2)$, and transitively on the set $\text{Gr}_{\mathbb{H}}(\mathbb{O}) \cong G_2/SO(4)$ of quaternion subalgebras [14, 20]. The stabilizer of an orthonormal pair (u, n) fixes the quaternion algebra $\mathbb{H}(u, n) = \mathbb{R} + \mathbb{R}u + \mathbb{R}n + \mathbb{R}un$ pointwise. Given $a, b \in \mathbb{O}$ there is $g \in G_2$ with $ga \in \mathbb{R} + \mathbb{R}e_1$ and $gb \in \mathbb{R} + \mathbb{R}e_1 + \mathbb{R}e_2$; we call this the *normal form* of the pair.

2.4. Quadratic algebras and the stretch operator. An algebra A is *quadratic* if every x satisfies $x^2 - 2t(x)x + N(x) = 0$ with $t(x), N(x) \in \mathbb{R}$, and *flexible* if $(xy)x = x(yx)$. In a flexible quadratic algebra with $\bar{x} = 2t(x) - x$ one has

$$(3) \quad L_{\bar{x}}L_x = N(x)\text{Id} + \text{alt}_x,$$

because $\bar{x}(xy) = (\bar{x}x)y - [\bar{x}, x, y] = N(x)y + [x, x, y]$. When the adjoint identity $\langle xy, z \rangle = \langle y, \bar{x}z \rangle$ holds, $L_{\bar{x}} = L_x^\top$ and alt_x is symmetric. We call $L_{\bar{x}}L_x/N(x)$ the *stretch operator* of x (it is the operator M_a of [5]); x is a zero divisor exactly when it has the eigenvalue 0, and $\dim \text{Ann}(x)$ is its multiplicity, where $\text{Ann}(x) = \ker L_x = \{y : xy = 0\}$.

3. THE MIRROR DOUBLE

Definition 3.1. For a $*$ -algebra A the *mirror double* $M(A)$ is $A \oplus A\ell$ with the product (2) and the involution $\overline{a + b\ell} = \bar{a} - b\ell$. The *mirror octonions* are $M(\mathbb{H})$ and the *mirror sedenions* are $\mathbb{S}' = M(\mathbb{O})$.

Proposition 3.2. Let A be a $*$ -algebra with $x\bar{x} = \bar{x}x = N(x) \in \mathbb{R}$ for all x . Then $M(A)$ is unital with unit 1, its involution is an anti-automorphism, $x\bar{x} = \bar{x}x = N(x) = N(a) + N(b)$, and $M(A)$ is quadratic with $x^2 = 2a_0x - N(x)$ where $a_0 = \text{Re } a$. If A is commutative then $M(A) = \text{CD}(A)$. The maps $\varepsilon: a + b\ell \mapsto a - b\ell$ and, for $\varphi \in \text{Aut}(A)$ commuting with the involution, $\varphi \oplus \varphi$, are automorphisms of $M(A)$.

Proof. Direct substitution in (2): $(a + b\ell)(\bar{a} - b\ell) = (\bar{a}a + \bar{b}b) + (-ba + b\bar{a})\ell = N(x)$, and symmetrically. For the quadratic identity, $(a + b\ell)^2 = (a^2 - \bar{b}b) + (ba + b\bar{a})\ell = (2a_0a - N(a) - N(b)) + 2a_0b\ell$. The rest is immediate from the shape of (2). $\square$

3.1. The embedding theorem.

Theorem 3.3. Let A be a $*$ -algebra, $\text{CD}(A) = A \oplus Ae$ and $\text{CD}^2(A) = \text{CD}(A) \oplus \text{CD}(A)e'$. The subspace $A \oplus (Ae)e'$ is a subalgebra of $\text{CD}^2(A)$, and

$$\Phi: M(A) \longrightarrow \text{CD}^2(A), \quad \Phi(a + b\ell) = \bar{a} + (be)e'$$

is an isomorphism of $*$ -algebras onto it. Moreover $(be)e' = \bar{b}(ee')$, so the subalgebra is $A + Aj$ with $j = ee'$, and in these coordinates

$$(a + bj)(c + dj) = (ac - \bar{b}d) + (ad + \bar{c}b)j.$$

Proof. Write elements of $\text{CD}(A)$ as pairs $(p, q) = p + qe$ and of $\text{CD}^2(A)$ as pairs of pairs. With $p = (a, 0), q = (0, b), r = (c, 0), s = (0, d)$ the product (1) in $\text{CD}^2(A)$ gives

$$(p, q)(r, s) = (pr - \bar{s}q, sp + q\bar{r}), \quad pr = (ac, 0), \quad \bar{s}q = (0, -d)(0, b) = (\bar{b}d, 0),$$

$$sp = (0, d)(a, 0) = (0, d\bar{a}), \quad q\bar{r} = (0, b)(\bar{c}, 0) = (0, bc),$$

so that

$$(4) \quad (a + (be)e')(c + (de)e') = (ac - \bar{b}d) + ((d\bar{a} + bc)e)e'.$$

Replacing a, c by $\bar{a}, \bar{c}$ in (4), the product of $\Phi(a + b\ell)$ and $\Phi(c + d\ell)$ is $(\bar{a}\bar{c} - \bar{b}d) + ((da + b\bar{c})e)e'$, while Φ of the mirror product (2) is $\bar{c}a - \bar{d}b + ((da + b\bar{c})e)e' = (\bar{a}\bar{c} - \bar{b}d) + ((da + b\bar{c})e)e'$. So Φ is multiplicative; it is visibly injective, linear, and compatible with the involutions. Finally $ee' = (e, 0)(0, 1) = (0, e)$ and $\bar{b}(ee') = (\bar{b}, 0)(0, e) = (0, e\bar{b})$; in $\text{CD}(A)$, $e\bar{b} = (0, 1)(\bar{b}, 0) = (0, b) = be$, so $\bar{b}(ee') = (0, be) = (be)e'$. Substituting $b \mapsto \bar{b}, d \mapsto \bar{d}$ in (4) and conjugating the j -coefficient gives the stated formula in the coordinates $a + bj$. $\square$

- Corollary 3.4.** (1) $M(\mathbb{H})$ is isomorphic to the subalgebra $\mathbb{H} + \mathbb{H}(\varepsilon\ell)$ of $\mathbb{S}$, for any quaternion subalgebra $\mathbb{H} \subset \mathbb{O}$ and unit $\varepsilon \perp \mathbb{H}$ in $\mathbb{O}$ (take $A = \mathbb{H}$, $e = \varepsilon$, $e' = \ell$; this is $\mathbb{H} + \mathbb{H}(\ell e)$ in the notation of [10]): this is the quasi-octonion algebra $\tilde{\mathbb{O}}$ of Cawagas [8] and the algebra S_8 of Chan–Đoković [10, Thm. 8.1], the unique conjugacy class of 8-dimensional subalgebras of $\mathbb{S}$ with zero divisors.
- (2) $\mathbb{S}'$ is isomorphic to the subalgebra $\mathbb{O} + \mathbb{O}(\ell e_{16})$ of $\mathbb{T} = \text{CD}(\mathbb{S})$ (take $A = \mathbb{O}$, $e = \ell$, $e' = e_{16}$), spanned by $e_0, \dots, e_7, e_{24}, \dots, e_{31}$: this is the algebra $\mathbb{S}^\gamma$ of Cawagas et al. [9].
- (3) $\mathbb{S}'$ is flexible and power-associative, satisfies the adjoint identity $\langle xy, z \rangle = \langle y, \bar{x}z \rangle = \langle x, z\bar{y} \rangle$, and $L_{\bar{x}}L_x = N(x) + \text{alt}_x$ with alt_x symmetric.

Proof. (1),(2) are Theorem 3.3 for $A = \mathbb{H}, \mathbb{O}$. (3): every Cayley–Dickson algebra is flexible and power-associative [22] and satisfies the adjoint identity $\langle xy, z \rangle = \langle y, \bar{x}z \rangle$ (see [18, 3]); Φ is an isometry, so these properties pass to the subalgebra. $\square$

Remark 3.5. The theorem explains the reversal in (2): $j = ee'$ behaves like a doubling unit ($ja = \bar{a}j$, $j^2 = -1$) but the associator $[a, e, e']$ does not vanish, and it is this associator that reverses every product in which two “ j -halves” meet. Iterating, $M(\mathbb{S}) \subset \text{CD}^3(\mathbb{O})$, and in general each of the 2^{n-3} words of length $n - 3$ in $\{\text{CD}, M\}$ applied to $\mathbb{O}$ produces a 2^n -dimensional subalgebra of $\text{CD}^{2(n-3)}(\mathbb{O})$.

3.2. Bales’s eight products give two algebras. Bales [2] shows that among thirty-two candidate doubling formulas $(a, b)(c, d) = (f, g)$, with f one of $ca - \bar{b}d$, $ca - d\bar{b}$, $ac - \bar{b}d$, $ac - d\bar{b}$, $ca - b\bar{d}$, $ca - d\bar{b}$, $ac - b\bar{d}$, $ac - d\bar{b}$ and g one of $\bar{d}a + bc$, $\bar{a}d + cb$, $ad + \bar{c}b$, $da + b\bar{c}$, exactly eight produce, at each level of a uniform tower, an algebra in which every basis triple $\{e_p, e_q, e_p e_q\}$ spans a quaternion algebra. They are

$$\begin{aligned} P_0 &= (ca - \bar{b}d, \bar{d}a + bc), & P_1 &= (ca - d\bar{b}, \bar{a}d + cb), \\ P_2 &= (ac - \bar{b}d, \bar{d}a + bc), & P_3 &= (ac - d\bar{b}, \bar{a}d + cb), \\ P_0^T &= (ca - b\bar{d}, ad + \bar{c}b), & P_1^T &= (ca - \bar{d}b, da + b\bar{c}), \\ P_2^T &= (ac - b\bar{d}, ad + \bar{c}b), & P_3^T &= (ac - \bar{d}b, da + b\bar{c}). \end{aligned}$$

P_3^T is (1) and P_1^T is (2).

Theorem 3.6. Let P be one of the eight products and let $P(\mathbb{O})$ denote $\mathbb{O} \oplus \mathbb{O}\ell$ with the product P , the octonions being multiplied in the standard $\mathbb{O}$. Then

$$P_0(\mathbb{O}) \cong P_3(\mathbb{O}) \cong P_0^T(\mathbb{O}) \cong P_3^T(\mathbb{O}) = \mathbb{S}, \quad P_1(\mathbb{O}) \cong P_2(\mathbb{O}) \cong P_2^T(\mathbb{O}) \cong P_1^T(\mathbb{O}) = \mathbb{S}',$$

with isomorphisms $a + b\ell \mapsto \bar{a} + b\ell$ ($P_3^T \rightarrow P_0$), $a + \bar{b}\ell$ ($P_3^T \rightarrow P_3$), $\bar{a} + \bar{b}\ell$ ($P_3^T \rightarrow P_0^T$), and $a + \bar{b}\ell$ ($P_1^T \rightarrow P_1$), $\bar{a} + b\ell$ ($P_1^T \rightarrow P_2$), $\bar{a} + \bar{b}\ell$ ($P_1^T \rightarrow P_2^T$). The two classes are not isomorphic. Each of the twenty-four remaining formulas, applied to $\mathbb{O}$, gives a unital algebra with $x\bar{x} = N(x)$, anticommuting basis units squaring to -1 and derivation algebra of dimension 14, but in every case some basis line $\{e_p, e_q, e_p e_q\}$ not contained in V fails to be a quaternion algebra; twenty of them have only the founding octave among their basis hyperplanes, and four have eight octonion hyperplanes (in the configuration of $\mathbb{S}'$) but are not graded-isomorphic to $\mathbb{S}$ or $\mathbb{S}'$.

Proof. Each isomorphism is checked by substituting into the formulas and using $\overline{xy} = \bar{y}\bar{x}$; for instance under $\psi(a + b\ell) = \bar{a} + b\ell$ the P_0 -product of $\psi(x), \psi(y)$ is $(\bar{c}\bar{a} - \bar{b}d) + (da + b\bar{c})\ell$ while ψ of the P_3^T -product $(ac - \bar{d}b) + (da + b\bar{c})\ell$ is $(\bar{c}\bar{a} - \bar{b}d) + (da + b\bar{c})\ell$. The other five are identical computations. Non-isomorphism of the two classes follows from Theorem 5.1 (the norm-one zero divisors of $\mathbb{S}$ form an 11-manifold [3], those of $\mathbb{S}'$ a 13-manifold), or from Theorem 7.1. The statements about the other twenty-four formulas are the computer census of Appendix A; the 14-dimensional derivation algebra is in every case the diagonal $\mathfrak{g}_2$. $\square$

Remark 3.7 (The orientation bit). In the list above, P_1 and P_3 differ only in the order of the first product, and likewise P_0/P_2 , P_0^T/P_2^T , P_1^T/P_3^T ; each such swap exchanges the two classes. Inside either class the choice of product is a gauge (an isomorphism of the simple form above), but between the classes it is a genuine bit: the orientation of the founding octave relative to the doubling twist. Since $\mathbb{O} \cong \mathbb{O}^{\text{op}}$ via conjugation, the bit is invisible on $\mathbb{O}$ alone and appears only when the two halves are coupled.

Remark 3.8 (Uniform towers). Bales applies each product at every level from $\mathbb{R}$ upward. For P_1^T this is not the mirror sedenions: the dimension-8 stage of the uniform P_1^T tower fails to be a composition algebra (it is $M(\mathbb{H})$, with 2-dimensional annihilators and Der of dimension 6), and its dimension-16 stage has Der of dimension 6, not 14. The algebra $\mathbb{S}'$ requires the mixed recipe: standard products up to $\mathbb{O}$, then (2). In the same vein, the standard double $\text{CD}(M(\mathbb{H}))$ of the quasi-octonions is again $\mathbb{S}$ (a signed relabelling of the sixteen basis units carries one to the other; Appendix A), while the double mirror $M(M(\mathbb{H}))$ has no octonion subalgebra among its basis hyperplanes and Der of dimension 6. Thus at dimension 16 the four words in $\{\text{CD}, M\}$ give exactly three algebras: $\mathbb{S}$ (twice), $\mathbb{S}'$, and $M(M(\mathbb{H}))$.

4. THE STRETCH OPERATOR

Theorem 4.1. *Let $x = a + b\ell \in \mathbb{S}'$, $N = N(x) = |a|^2 + |b|^2$, and let $b_{\parallel}$ be the orthogonal projection of b onto $\mathbb{C}_a = \mathbb{R} + \mathbb{R}a'$. The symmetric operator $\text{alt}_x = L_{\bar{x}}L_x - N$ has eigenvalues*

$$\pm 2|a'| |b| \text{ (multiplicity 2 each),} \quad \pm 2|a'| |b_{\parallel}| \text{ (multiplicity 4 each),} \quad 0 \text{ (multiplicity 4).}$$

Equivalently, the stretch operator $L_{\bar{x}}L_x/N$ has spectrum $\{1 \pm \sigma, 1 \pm \tau, 1\}$ with multiplicities 2, 2, 4, 4, 4 where

$$\sigma = \frac{2|a'| |b|}{|a|^2 + |b|^2}, \quad \tau = \frac{2|a'| |b_{\parallel}|}{|a|^2 + |b|^2}, \quad 0 \leq \tau \leq \sigma \leq 1.$$

If $a' \neq 0$ and $b_{\parallel} \neq 0$ then $\ker \text{alt}_x = \mathbb{C}_a + (b\mathbb{C}_a)\ell$, a quaternion subalgebra of $\mathbb{S}'$ containing x .

Proof. By Artin's theorem a and b lie in a quaternion subalgebra $Q \subset \mathbb{O}$; write $\mathbb{O} = Q \oplus Q\varepsilon$ with $\varepsilon \perp Q$ a unit, so $\mathbb{O} = \text{CD}(Q)$ via (1). Then $\mathbb{S}' = M(Q) \oplus M(Q)^{\perp}$ with $M(Q) = Q + Q\ell$ and $M(Q)^{\perp} = Q\varepsilon + (Q\varepsilon)\ell$. We claim L_x preserves both summands. On $M(Q)$ this is because $M(Q)$ is a subalgebra. For $c, d \in Q$, using (1) in $\mathbb{O} = \text{CD}(Q)$, $(c\varepsilon)a = (c\bar{a})\varepsilon$, $(d\varepsilon)b = (d\bar{b})\varepsilon$, $\bar{c}\varepsilon = -c\varepsilon$, and $b(c\varepsilon) = (cb)\varepsilon$, so (2) gives

$$(5) \quad x(c\varepsilon + (d\varepsilon)\ell) = (c\bar{a} + d\bar{b})\varepsilon + ((d\bar{a} - cb)\varepsilon)\ell,$$

which lies in $M(Q)^{\perp}$. Hence alt_x is block diagonal, and we compute the two blocks.

The block on $M(Q)^{\perp}$. By (5), in the coordinates $(c, d) \in Q^2$, L_x is right multiplication by the quaternionic matrix $X = \begin{pmatrix} \bar{a} & -b \\ b & \bar{a} \end{pmatrix}$ (row vectors, $(c, d) \mapsto (c\bar{a} + d\bar{b}, -cb + d\bar{a})$), and $L_{\bar{x}}$ is right multiplication by $X' = \begin{pmatrix} a & b \\ -\bar{b} & a \end{pmatrix}$. Since Q is associative,

$$XX' = \begin{pmatrix} N & \bar{a}b - ba \\ \bar{b}a - a\bar{b} & N \end{pmatrix} = \begin{pmatrix} N & w \\ \bar{w} & N \end{pmatrix}, \quad w := \bar{a}b - ba = 2(\langle a, b \rangle - b_0 a),$$

so on this block $\text{alt}_x(c, d) = (d\bar{w}, cw)$ and $\text{alt}_x^2 = |w|^2$. Thus alt_x has eigenvalues $\pm |w|$ there, each with multiplicity 4: it is symmetric, and it exchanges the two slots $Q\varepsilon$ and $(Q\varepsilon)\ell$, so its trace on the block vanishes. Finally $|w|^2 = 4(\langle a, b \rangle^2 - 2a_0b_0\langle a, b \rangle + b_0^2|a|^2) = 4(\langle a', b' \rangle^2 + b_0^2|a'|^2) = 4|a'|^2|b_{\parallel}|^2$, since $|b_{\parallel}|^2 = b_0^2 + \langle b', a' \rangle^2/|a'|^2$.

The block on $M(Q)$. For $c, d \in Q$ put $y = c + d\ell$. From (2) with Q associative,

$$xy = (ca - \bar{d}b) + (da + b\bar{c})\ell, \quad x(xy) = (ca^2 - \bar{d}ba - \bar{a}\bar{d}b - |b|^2c) + (da^2 + b\bar{c}a + b\bar{a}\bar{c} - |b|^2d)\ell,$$

and $x^2y = (x^2)y$ with $x^2 = (a^2 - |b|^2) + 2a_0b\ell$ gives $x^2y = (ca^2 - |b|^2c - 2a_0\bar{d}b) + (da^2 - |b|^2d + 2a_0b\bar{c})\ell$. Subtracting and using $\bar{a} - 2a_0 = -a$,

$$(6) \quad \text{alt}_x(c + d\ell) = [\bar{d}b, a] + (b[\bar{c}, \bar{a}])\ell,$$

where $[p, q] = pq - qp$. Applying (6) twice and using $\overline{[\bar{c}, \bar{a}]} = [a, c]$, $\overline{[\bar{d}b, a]} = [\bar{a}, \bar{b}d]$,

$$\text{alt}_x^2(c + d\ell) = |b|^2 [[a, c], a] + (b [[\bar{a}, \bar{b}d], \bar{a}])\ell.$$

In $Q \cong \mathbb{H}$, $[[a, q], a] = -\text{ad}_a^2(q) = 4|a'|^2 q'_\perp$, where $q'_\perp$ is the component of q orthogonal to $\mathbb{C}_a$; the same holds for $\bar{a}$. Hence alt_x^2 acts on the c -slot as $4|a'|^2|b|^2$ times the orthogonal projection onto $\mathbb{C}_a^\perp \cap Q$, and on the d -slot as $4|a'|^2|b|^2$ times the orthogonal projection onto $b(\mathbb{C}_a^\perp \cap Q)$ (the map $d \mapsto \bar{b}d$ is $|b|$ times an isometry). So on $M(Q)$ the operator alt_x^2 has eigenvalue $4|a'|^2|b|^2$ with multiplicity 4 and 0 with multiplicity 4; by symmetry and vanishing trace ((6) again exchanges the two slots), alt_x has eigenvalues $\pm 2|a'||b|$ with multiplicity 2 each and 0 with multiplicity 4, with kernel $\mathbb{C}_a + (b\mathbb{C}_a)\ell$ whenever $|a'||b| \neq 0$.

Combining the blocks gives the spectrum. The inequalities $\tau \leq \sigma \leq 1$ are $|b_\parallel| \leq |b|$ and $2|a'||b| \leq |a|^2 + |b|^2$. When $|w| \neq 0$ the kernel of alt_x is the kernel on $M(Q)$, namely $\mathbb{C}_a + (b\mathbb{C}_a)\ell$; identifying $z + (bz')\ell \leftrightarrow (z, z')$ for $z, z' \in \mathbb{C}_a$ and $|b| = 1$, the product (2) becomes $(z_1z_2 - \bar{z}'_2z'_1) + (z'_2z_1 + z'_1\bar{z}_2)\ell$, which is $\text{CD}(\mathbb{C}_a) \cong \mathbb{H}$. $\square$

Remark 4.2. In the normal form $a = a_0 + a_1e_1$, $b = b_0 + b_1e_1 + b_2e_2$ the characteristic polynomial of alt_x is

$$\lambda^4 (4a_1^2(b_0^2 + b_1^2) - \lambda^2)^4 (4a_1^2(b_0^2 + b_1^2 + b_2^2) - \lambda^2)^2,$$

as computer algebra confirms directly (Appendix A). For $\mathbb{S}$ the same computation gives $\lambda^8(4a_1^2b_2^2 - \lambda^2)^4$, i.e. eigenvalues $\pm 2|a' \times b'|$ of multiplicity 4 and 0 of multiplicity 8, recovering [5, Thm. 7.2] and extending it to non-imaginary a, b . Thus the “private” subalgebra $\ker \text{alt}_x$ of a generic element is a quaternion algebra in $\mathbb{S}'$ and an octonion algebra in $\mathbb{S}$ (for $\mathbb{S}$ the 8-dimensional kernel is again a composition subalgebra; Appendix A).

Corollary 4.3 (Alternative elements). *$x = a + b\ell$ satisfies $[x, x, y] = 0$ for all y if and only if $a \in \mathbb{R}$ or $b = 0$. Thus the set of alternative elements of $\mathbb{S}'$ is the union $\mathbb{O} \cup (\mathbb{R} + \mathbb{O}\ell)$ of two 8-dimensional subspaces meeting in $\mathbb{R}$, exactly one of which, $\mathbb{O}$, is a subalgebra. In $\mathbb{S}$ the alternative elements are $\{a + b\ell : a' \times b' = 0\} = (\mathbb{R} + \mathbb{O}\ell) \cup \bigcup_{u \in S^6} (\mathbb{C}_u + \mathbb{C}_u\ell)$.*

Proof. $\text{alt}_x = 0$ iff all eigenvalues vanish iff $|a'||b| = 0$ (then automatically $|a'||b_\parallel| = 0$). The statement for $\mathbb{S}$ is the analogous reading of $\lambda^8(4a_1^2b_2^2 - \lambda^2)^4$. $\square$

5. ZERO DIVISORS

5.1. The dead set.

Theorem 5.1. *$x = a + b\ell \in \mathbb{S}'$ is a zero divisor if and only if $\text{Re } a = 0$ and $|a| = |b| \neq 0$. The set of zero divisors of norm one is*

$$\mathcal{Z}(\mathbb{S}') = \left\{ u + b\ell : u \in \text{Im } \mathbb{O}, |u| = |b| = \frac{1}{\sqrt{2}} \right\} = S^6\left(\frac{1}{\sqrt{2}}\right) \times S^7\left(\frac{1}{\sqrt{2}}\right),$$

a 13-dimensional Riemannian product of round spheres in the induced metric, and the cone over it is the full set of zero divisors. Left and right zero divisors coincide, and $xy = 0$ implies $yx = 0$.

Proof. By Theorem 4.1, 0 is an eigenvalue of $L_{\bar{x}}L_x = L_x^\top L_x$ iff $N = 2|a'||b|$ (since $\tau \leq \sigma$), i.e. $|a|^2 + |b|^2 = 2|a'||b|$, i.e. $a_0^2 + (|a'| - |b|)^2 = 0$. The set $\{|a| = |b| = 1/\sqrt{2}, a_0 = 0\} \subset \mathbb{O} \times \mathbb{O}$ is the product of the two spheres, and the inner product of $\mathbb{R}^{16} = \mathbb{O} \times \mathbb{O}$ is the product metric. Since $\bar{x}$ is again of the form $u' + b'\ell$ with u' imaginary and $|u'| = |b'|$, and $yx = 0 \iff \bar{x}\bar{y} = 0$, left and right zero divisors coincide. Two-sidedness of annihilation is verified after Theorem 5.2. $\square$

We give a second, direct proof of necessity, because it isolates the mechanism. Suppose $(a + b\ell)(c + d\ell) = 0$ with $x \neq 0$, i.e.

$$(7) \quad ca = \bar{d}b, \quad da = -b\bar{c}.$$

Taking norms, $|c||a| = |d||b|$ and $|d||a| = |b||c|$; if $y \neq 0$ then $c, d \neq 0$ and $|a| = |b|$, and $a \neq 0$. From the first equation $c = (\bar{d}b)a^{-1}$ and $\bar{c} = (a(\bar{b}d))/|a|^2$; inserting into the second, $|a|^2 da = -b(a(\bar{b}d))$. Pair both sides with d using the adjoint identity: $\langle da, d \rangle = |d|^2 a_0$ and $\langle b(a(\bar{b}d)), d \rangle = \langle a(\bar{b}d), \bar{b}d \rangle = |b|^2 |d|^2 a_0$. Hence $|d|^2 a_0(|a|^2 + |b|^2) = 0$ and $a_0 = 0$.

5.2. Annihilators.

Theorem 5.2. *Let $x = u + b\ell$ with $u \in \text{Im } \mathbb{O}$, $|u| = |b| \neq 0$. Then*

$$\text{Ann}(x) = \{ nu + (bn)\ell : n \in \text{Im } \mathbb{O}, n \perp u, [b, n, u] = 0 \}.$$

Consequently $\dim \text{Ann}(x) = 6$ if $b \in \mathbb{C}_u$ and $\dim \text{Ann}(x) = 2$ otherwise; in the latter case the admissible n form the plane $\mathbb{R}b_\perp + \mathbb{R}ub_\perp$, where $b_\perp = b - b_\parallel$, and $\text{Ann}(x)$ is the plane spanned by $b_\perp u + (bb_\perp)\ell$ and $(ub_\perp)u + (b(ub_\perp))\ell$.

Proof. By homogeneity assume $|u| = |b| = 1$. Given $y = c + d\ell$ put $n = \bar{b}d$, so $d = bn$. The first equation of (7) reads $cu = \bar{d}b = (\bar{n}\bar{b})b = \bar{n}$, i.e. $c = -\bar{n}u$; conversely this c satisfies it. With $c = -\bar{n}u$ we have $\bar{c} = un$ and the second equation of (7) becomes

$$(8) \quad (bn)u + b(un) = 0.$$

Pair (8) with b : $\langle (bn)u, b \rangle = \langle bn, b\bar{u} \rangle = \langle n, \bar{u} \rangle = -\langle n', u \rangle$ and $\langle b(un), b \rangle = \langle un, 1 \rangle = -\langle n', u \rangle$, so $n' \perp u$. Pair it with bu : $\langle (bn)u, bu \rangle = \langle bn, b \rangle = n_0$ and $\langle b(un), bu \rangle = \langle un, u \rangle = n_0$, so $n_0 = 0$. Thus $n \in \text{Im } \mathbb{O}$, $n \perp u$, hence $un = -nu$ and $\bar{n} = -n$, and (8) becomes $(bn)u - b(nu) = [b, n, u] = 0$. Conversely, if $n \in \text{Im } \mathbb{O}$, $n \perp u$ and $[b, n, u] = 0$, then $c = nu$, $d = bn$ satisfy (7). This proves the description. If $b = \beta_0 + \beta_1 u \in \mathbb{C}_u$ then $[b, n, u] = \beta_1[u, n, u] = 0$ for every n by flexibility, giving the 6-dimensional space $\{n \in \text{Im } \mathbb{O} : n \perp u\}$. If $b \notin \mathbb{C}_u$, Lemma 2.2 says $n \in \mathbb{H}(u, b) = \mathbb{C}_u + \mathbb{C}_u b_\perp$, and $n \in \text{Im } \mathbb{O}$, $n \perp u$ forces $n \in \mathbb{R}b_\perp + \mathbb{R}ub_\perp$. $\square$

Remark 5.3 (Two-sidedness). For $y = nu + (bn)\ell \in \text{Ann}(x)$ one computes $yx = (u(nu) - \bar{b}(bn)) + (b(nu) - (bn)u)\ell = (n - n) - [b, n, u]\ell = 0$, using $u(nu) = -(uu)n = n$ for $n \perp u$. So annihilation in $\mathbb{S}'$ is two-sided, as in $\mathbb{S}$.

Proposition 5.4 (Basis zero divisors). *For $1 \leq i \leq 7$ and $8 \leq j \leq 15$, both $e_i + e_j$ and $e_i - e_j$ are zero divisors of $\mathbb{S}'$: there are 56 such index pairs and 112 two-term zero divisors, against de Marrais' 42 "assessors" and 84 two-term zero divisors of $\mathbb{S}$ [11], which are the pairs with $j \neq i + 8$. Both algebras have exactly 336 ordered pairs $(e_i \pm e_j, e_k \pm e_l)$ of two-term elements with product zero. In the conventions of 2.1:*

$$\begin{aligned} \text{Ann}(e_1 + e_9) &= \text{span}\{e_k - e_{k+8} : 2 \leq k \leq 7\}, \\ (e_1 + e_8)(e_{10} - e_3) &= 0, \quad (e_1 + e_{10})(e_2 + e_9) = 0 = (e_1 + e_{10})(e_3 + e_8). \end{aligned}$$

Proof. $e_i + e_j = e_i + e_{j-8}\ell$ has imaginary first component and equal norms, so Theorem 5.1 applies; the annihilators are Theorem 5.2 with $u = e_i$, $b = e_{j-8}$ (for $e_1 + e_9$: $b = u$, n ranges over $e_2, \dots, e_7$, and $nu + (un)\ell = nu - (nu)\ell$). The counts are a finite check (Appendix A). $\square$

In the colour vocabulary of [16], where $e_1, \dots, e_7$ are *hues* and e_{i+8} is the *ultraviolet* of e_i : in $\mathbb{S}'$ a hue with its own ultraviolet, or with the bare ultraviolet ℓ , is not innocent, and it is killed by every other hue minus that hue's ultraviolet.

5.3. The manifold of zero-divisor pairs. Let $\mathcal{P}(\mathbb{S}') = \{(x, y) \in S^{15} \times S^{15} : xy = 0\}$.

Theorem 5.5. *The map*

$$\Psi: V_2(\mathbb{R}^7) \times S^3 \longrightarrow \mathcal{P}(\mathbb{S}'), \quad \Psi(u, n, q) = \left(\frac{u+b\ell}{\sqrt{2}}, \frac{nu+(bn)\ell}{\sqrt{2}} \right), \quad b = q_0 + q_1u + q_2n + q_3un,$$

is a diffeomorphism. Hence $\mathcal{P}(\mathbb{S}')$ is a smooth closed 14-manifold, the trivial S^3 -bundle over the Stiefel manifold. The diagonal action of G_2 corresponds to its action on $V_2(\mathbb{R}^7)$ alone: every stabilizer is $SU(2)$, the orbits are copies of $V_2(\mathbb{R}^7)$, and the orbit space is S^3 . The projection $\mathcal{P}(\mathbb{S}') \rightarrow \mathcal{Z}(\mathbb{S}')$, $(x, y) \mapsto x$, is surjective with fibre S^1 over $u + b\ell$ with $b \notin \mathbb{C}_u$ and fibre S^5 over the 7-dimensional locus $b \in \mathbb{C}_u$. The map $(x, y) \mapsto \mathbb{H}(u, n)$ is a G_2 -equivariant fibre bundle $\mathcal{P}(\mathbb{S}') \rightarrow \text{Gr}_{\mathbb{H}}(\mathbb{O}) = G_2/SO(4)$ with fibre $V_2(\mathbb{R}^3) \times S^3 \cong SO(3) \times S^3$. Finally, $\mathcal{P}(\mathbb{S}')$ is not homotopy equivalent to $G_2 \cong \mathcal{P}(\mathbb{S})$.

Proof. By Theorems 5.1 and 5.2, $(x, y) \in \mathcal{P}(\mathbb{S}')$ iff $x = (u + b\ell)/\sqrt{2}$ with $u \in S^6$, $b \in S^7$, and $y = (nu + (bn)\ell)/\sqrt{2}$ with $n \in \text{Im } \mathbb{O}$, $|n| = 1$ (because $|nu| = |bn| = |n|$), $n \perp u$ and $[b, n, u] = 0$. For an orthonormal pair (u, n) the last condition reads $[n, b, u] = 0$ (the associator is alternating), and by Lemma 2.2 applied to the pair (n, u) with b as the variable it is $b \in \mathbb{H}(u, n) = \mathbb{R} + \mathbb{R}u + \mathbb{R}n + \mathbb{R}un$, i.e. $b = q_0 + q_1u + q_2n + q_3un$ with $q \in S^3$. So Ψ is a bijection. Its inverse $(x, y) \mapsto (u, n, q)$ with $u = \sqrt{2}x_{\mathbb{O}}$, $b = \sqrt{2}x_{\mathbb{O}\ell}$, $n = -(\sqrt{2}y_{\mathbb{O}})u$, $q = (\langle b, 1 \rangle, \langle b, u \rangle, \langle b, n \rangle, \langle b, un \rangle)$, is the restriction of a smooth map on $\mathbb{R}^{32}$; hence Ψ is a smooth embedding and $\mathcal{P}(\mathbb{S}')$ an embedded submanifold diffeomorphic to $V_2(\mathbb{R}^7) \times S^3$. For $g \in G_2$, $g(un) = (gu)(gn)$, so $g\Psi(u, n, q) = \Psi(gu, gn, q)$. The stabilizer of (u, n) in G_2 is $SU(2)$ and G_2 acts transitively on $V_2(\mathbb{R}^7)$; the invariant q parametrizes the orbits. The fibre of $(x, y) \mapsto x = (u, b)$ is $\{n \in S_u^5 : b \in \mathbb{H}(u, n)\}$: all of $S_u^5 = \{n \in \text{Im } \mathbb{O} : |n| = 1, n \perp u\}$ if $b \in \mathbb{C}_u$, and the unit circle of $\mathbb{R}b_{\perp} + \mathbb{R}ub_{\perp}$ otherwise. The map to $\text{Gr}_{\mathbb{H}}(\mathbb{O})$ is G_2 -equivariant onto a homogeneous space, hence a fibre bundle; its fibre over Q is $\{(u, n, q) : u, n \in \text{Im } Q\} = V_2(\text{Im } Q) \times S^3$, and $V_2(\mathbb{R}^3) \cong SO(3)$.

For the last claim, $H^*(G_2; \mathbb{F}_2) = \mathbb{F}_2[x_3]/(x_3^4) \otimes \Lambda(x_5)$ [6, 17], so the cup square $H^3 \rightarrow H^6$ is nonzero. On the other hand $V_2(\mathbb{R}^7)$ is the unit tangent bundle of S^6 , whose Euler class is 2, so $H^*(V_2(\mathbb{R}^7); \mathbb{F}_2) = \Lambda(x_5, x_6)$ and $H^*(V_2(\mathbb{R}^7) \times S^3; \mathbb{F}_2) = \Lambda(x_5, x_6) \otimes \Lambda(y_3)$, in which $y_3^2 = 0$. Both H^3 are one-dimensional, so the rings are not isomorphic. $\square$

Remark 5.6 (Comparison with the sedenions). For $\mathbb{S}$, $\mathcal{Z}(\mathbb{S}) = \{u + v\ell : u, v \in \text{Im } \mathbb{O}, u \perp v, |u| = |v| = 1/\sqrt{2}\} \cong V_2(\mathbb{R}^7)$ is an 11-manifold [3, Thm. 1.3], all annihilators are 4-dimensional ([18, Cor. 1.17] gives $\dim \text{Ann} \equiv 0 \pmod{4}$; exactness is [3, Prop. 12.1]), and $\mathcal{P}(\mathbb{S}) \cong G_2$ [18, Cor. 2.14], the principal $SU(2)$ -bundle $G_2 \rightarrow G_2/SU(2) = V_2(\mathbb{R}^7)$. Reggiani [19] shows that $\mathcal{P}(\mathbb{S})$ with the induced metric is isometric to G_2 with a naturally reductive left-invariant metric and fibres over $G_2/SO(4)$ with fibres locally isometric to $S^3 \times S^3$; the fibre $SO(4) = (S^3 \times S^3)/\{\pm 1\}$ of $\mathcal{P}(\mathbb{S})$ and the fibre $SO(3) \times S^3 = (S^3/\{\pm 1\}) \times S^3$ of $\mathcal{P}(\mathbb{S}')$ over the same Grassmannian are two different $\mathbb{Z}/2$ -quotients of $S^3 \times S^3$. Table 1 collects the comparison.

6. OCTAVES

The basis e_g , $g \in \mathbb{F}_2^4$, of $\mathbb{S}'$ satisfies $e_g e_h = \pm e_{g \oplus h}$, so every linear subspace of $\mathbb{F}_2^4$ spans a subalgebra. The fifteen hyperplanes of $\mathbb{F}_2^4$ are the kernels of the nonzero functionals $f \in (\mathbb{F}_2^4)^*$; identifying $(\mathbb{F}_2^4)^*$ with $\mathbb{F}_2^4$ by the dot product, the hyperplane $H_f = f^{\perp}$ contains the doubling point $\ell = 8 = (0, 0, 0, 1)$ iff $f_4 = 0$. Write $V = \mathbb{F}_2^3 \times 0$ for the founding octave ($f = 8$).

Theorem 6.1. *Let $H \subset \mathbb{F}_2^4$ be a hyperplane and let $A_H \subset \mathbb{S}'$, $B_H \subset \mathbb{S}$ be the 8-dimensional subalgebras it spans.*

	$\mathbb{S} = \text{CD}(\mathbb{O})$	$\mathbb{S}' = \text{M}(\mathbb{O})$
norm-one zero divisors $\mathcal{Z}$	$V_2(\mathbb{R}^7)$, dim 11	$S^6 \times S^7$, dim 13
condition on $x = a + b\ell$	$a, b \in \text{Im } \mathbb{O}$, $a \perp b$, $ a = b $	$a \in \text{Im } \mathbb{O}$, $ a = b $
dim $\text{Ann}(x)$	4	2; 6 when $b \in \mathbb{C}_a$
pair manifold $\mathcal{P}$	G_2 (a G_2 -torsor)	$V_2(\mathbb{R}^7) \times S^3$ (stabilizers $SU(2)$)
$\mathcal{P} \rightarrow G_2/SO(4)$, fibre	$SO(4)$	$SO(3) \times S^3$
stretch spectrum, multiplicities	$1 \pm s$ (4, 4), 1 (8)	$1 \pm \sigma$ (2, 2), $1 \pm \tau$ (4, 4), 1 (4)
dim $\ker \text{alt}_x$, generic x	8 (an octonion subalgebra)	4 (a quaternion subalgebra)
alternative elements	$(\mathbb{R} + \mathbb{O}\ell) \cup \bigcup_u (\mathbb{C}_u + \mathbb{C}_u\ell)$	$\mathbb{O} \cup (\mathbb{R} + \mathbb{O}\ell)$
octonion basis hyperplanes	$\mathbb{O}$ and the 7 through ℓ	the 8 avoiding ℓ
quasi-octonion basis hyperplanes	the 7 avoiding ℓ except $\mathbb{O}$	the 7 through ℓ
two-term basis zero divisors	84 (de Marrais)	112
Der	$\mathfrak{g}_2$	$\mathfrak{g}_2$
Aut	$G_2 \times S_3$ (Brown)	$G_2 \times \mathbb{Z}/2$
graded automorphisms	order 2688 = $168 \cdot 16$ [15]	order 2688
inside $\mathbb{T} = \text{CD}(\mathbb{S})$	$\mathbb{S}$ (16 basis hyperplanes)	$\mathbb{S}^\gamma = \mathbb{O} + \mathbb{O}(ee')$ (1 hyperplane)

TABLE 1. The sedenions and the mirror sedenions. Here $s = 2|a' \times b'|/N$, $\sigma = 2|a'||b|/N$, $\tau = 2|a'||b_\parallel|/N$.

- (1) A_H is a composition algebra, hence isomorphic to $\mathbb{O}$, if and only if $\ell \notin H$. These are V and the seven hyperplanes $H_L^- = L \cup (L^\perp + \ell)$, L a line of the Fano plane $V \setminus 0$, spanning $\mathbb{H}_L + \mathbb{H}_L^\perp \ell$.
- (2) B_H is a composition algebra if and only if $H = V$ or $\ell \in H$; the latter are the seven $H_L^+ = L \cup (L + \ell)$, spanning $\mathbb{H}_L + \mathbb{H}_L \ell$.
- (3) The remaining seven subalgebras, $\mathbb{H}_L + \mathbb{H}_L \ell \subset \mathbb{S}'$ and $\mathbb{H}_L + \mathbb{H}_L^\perp \ell \subset \mathbb{S}$, are isomorphic to the mirror octonions $\text{M}(\mathbb{H})$, contain zero divisors, and have derivation algebras of dimension 6.

Proof. (1) V spans $\mathbb{O}^{\text{op}} \cong \mathbb{O}$. Let L be a line, $\mathbb{H} = \mathbb{H}_L$, and $\mathbb{H}^\perp = \mathbb{H}\varepsilon$ for a unit $\varepsilon \perp \mathbb{H}$. For $a, c \in \mathbb{H}$ and $b\varepsilon, d\varepsilon \in \mathbb{H}^\perp$ the mirror product is, using (1) in $\mathbb{O} = \text{CD}(\mathbb{H})$,

$$(a + (b\varepsilon)\ell)(c + (d\varepsilon)\ell) = (ca - \overline{d\varepsilon}(b\varepsilon)) + ((d\varepsilon)a + (b\varepsilon)\bar{c})\ell = (ca - \bar{b}d) + ((\bar{d}a + bc)\varepsilon)\ell.$$

Its norm is $|ca - \bar{b}d|^2 + |\bar{d}a + bc|^2 = (|a|^2 + |b|^2)(|c|^2 + |d|^2) - 2\text{Re}(ca\bar{d}b) + 2\text{Re}(\bar{d}a\bar{c}b)$, and in the associative $\mathbb{H}$, $\text{Re}(\bar{d}a\bar{c}b) = \text{Re}(\bar{b}cad) = \text{Re}(bca\bar{d}) = \text{Re}(\bar{d}bca) = \text{Re}(ca\bar{d}b)$ by cyclicity of Re ; hence the norm is multiplicative and $A_{H_L^-}$ is an 8-dimensional composition algebra, i.e. $\mathbb{O}$. For $\mathbb{H}_L + \mathbb{H}_L \ell$ the mirror product is that of $\text{M}(\mathbb{H}_L)$, and $\text{M}(\mathbb{H})$ has zero divisors by Theorem 8.1 below. (2) $\mathbb{H}_L + \mathbb{H}_L \ell \subset \mathbb{S}$ is $\text{CD}(\mathbb{H}_L) = \mathbb{O}$; and $\mathbb{H}_L + \mathbb{H}_L^\perp \ell \subset \mathbb{S}$ is the quasi-octonion algebra of Corollary 3.4(1), which has zero divisors. (3) follows from Theorem 8.1. $\square$

Thus the two algebras share the founding octave and exchange the roles of the seven hyperplanes through ℓ and the seven avoiding it: in $PG(3, 2)$, the octonion planes of $\mathbb{S}'$ are “all planes avoiding a point”, those of $\mathbb{S}$ “all planes through a point, plus one more”. The two configurations are not projectively equivalent: ℓ is the unique nonzero point on none of the eight hyperplanes avoiding it, so the stabilizer in $GL(4, 2)$ of $\mathbb{S}'$'s configuration is the full stabilizer of the point ℓ (order 1344), which permutes the eight hyperplanes transitively; the stabilizer of $\mathbb{S}$'s configuration fixes both ℓ (the intersection of the seven) and V (the odd one out) and has order 168. Nevertheless the graded automorphism groups coincide (Proposition 7.4): the transvections moving V to another H_L^- do not preserve the signs.

Proposition 6.2. *Let $(x, y) = \Psi(u, n, q) \in \mathcal{P}(\mathbb{S}')$. Then x and y lie in the quasi-octonion subalgebra $M(Q) = Q + Q\ell$, $Q = \mathbb{H}(u, n)$. In contrast, for a zero-divisor pair $x = a + b\ell$, $y = c + d\ell$ of $\mathbb{S}$ the octonions a, b, c, d generate all of $\mathbb{O}$: every zero-divisor pair of $\mathbb{S}$ is special [18, Cor. 2.12], and for special pairs the span is an octonion algebra [18, Thm. 2.13].*

Proof. $u, n \in Q$, $b \in Q$ by Theorem 5.5, hence $nu, bn \in Q$. $\square$

7. SYMMETRY

Theorem 7.1. $\text{Aut}(\mathbb{S}') = \{\varphi \oplus \pm\varphi : \varphi \in G_2\} \cong G_2 \times \mathbb{Z}/2$ and $\text{Der}(\mathbb{S}') = \{D \oplus D : D \in \mathfrak{g}_2\} \cong \mathfrak{g}_2$.

Proof. The maps listed are automorphisms by Proposition 3.2. Conversely let $\theta \in \text{Aut}(\mathbb{S}')$. Since $\mathbb{S}'$ is quadratic and for $x \notin \mathbb{R}$ the polynomial $X^2 - 2t(x)X + N(x)$ is the minimal polynomial of x , θ preserves t and N , hence the involution $\bar{x} = 2t(x) - x$ and the inner product. It also preserves the set Alt of alternative elements, which by Corollary 4.3 is the union of the subspaces $\mathbb{O}$ and $\mathbb{R} + \mathbb{O}\ell$. A linear subspace contained in a union of two subspaces lies in one of them, so θ permutes the two maximal subspaces of Alt ; since $\mathbb{O}$ is a subalgebra and $\mathbb{R} + \mathbb{O}\ell$ is not $((e_1\ell)(e_2\ell) = e_2e_1 \notin \mathbb{R} + \mathbb{O}\ell)$, $\theta(\mathbb{O}) = \mathbb{O}$ and hence $\theta(\mathbb{O}\ell) = \mathbb{O}\ell$. Write $\theta(a + b\ell) = \varphi(a) + \psi(b)\ell$. Restricting to $\mathbb{O}$ (with the opposite product) gives $\varphi \in \text{Aut}(\mathbb{O}^{\text{op}}) = \text{Aut}(\mathbb{O}) = G_2$. From $a(d\ell) = (da)\ell$ we get $\psi(da) = \psi(d)\varphi(a)$; with $d = 1$, $\psi(a) = q\varphi(a)$ where $q = \psi(1)$, $|q| = 1$; then $q\varphi(da) = (q\varphi(d))\varphi(a)$ for all a, d , i.e. $q(yz) = (qy)z$ for all $y, z \in \mathbb{O}$, which forces $q \in \mathbb{R}$ since the nucleus of $\mathbb{O}$ is $\mathbb{R}$. So $q = \pm 1$ and $\theta = \varphi \oplus \pm\varphi$. The statement on derivations follows because $\text{Der}(\mathbb{S}')$ is the Lie algebra of the Lie group $\text{Aut}(\mathbb{S}')$. $\square$

Remark 7.2. For $\mathbb{S}$, Brown [7] proved $\text{Aut}(\mathbb{S}) = G_2 \times S_3$; a generator of order three is the simultaneous rotation by $2\pi/3$ in each plane $\mathbb{R}q + \mathbb{R}q\ell$, $q \in \text{Im } \mathbb{O}$ (fixing 1 and ℓ) [12]. This map is not an automorphism of $\mathbb{S}'$ (a direct check, Appendix A), consistent with Theorem 7.1: it moves $\mathbb{O}$ to another alternative subspace, and in $\mathbb{S}'$ there is none to move it to. The two images of $\mathbb{O}$ under the order-three element are octonion subalgebras of $\mathbb{S}$ consisting of alternative elements; in $\mathbb{S}'$ the only 8-dimensional subalgebra consisting of alternative elements is $\mathbb{O}$ itself.

7.1. Graded automorphisms and the sign function. Both algebras are *twisted group algebras* of $G = \mathbb{F}_2^4$: $e_g e_h = F(g, h)e_{g \oplus h}$ with $F: G \times G \rightarrow \{\pm 1\}$. A *graded isomorphism* $\mathbb{R}_F[G] \rightarrow \mathbb{R}_{F'}[G]$ is a map $e_g \mapsto \lambda(g)e_{\sigma g}$ with $\sigma \in GL(4, 2)$ and $\lambda: G \rightarrow \{\pm 1\}$; it exists iff $F'(\sigma g, \sigma h)\lambda(g \oplus h) = F(g, h)\lambda(g)\lambda(h)$ for all g, h , a linear system over $\mathbb{F}_2$ for λ once σ is fixed.

Proposition 7.3. *Let $F_{\mathbb{S}}, F_{\mathbb{S}'}$ be the sign functions of $\mathbb{S}, \mathbb{S}'$ in the basis of Convention 2.1. Then $F_{\mathbb{S}'} = F_{\mathbb{S}} \cdot \chi$ where $\chi(g, h) = -1$ if g, h are distinct nonzero elements of $V = \mathbb{F}_2^3 \times 0$ and $\chi(g, h) = 1$ otherwise. The function χ is not a 2-cocycle: for $g \neq h$ nonzero in V and $k \notin V$, $\chi(g, h)\chi(g \oplus h, k) = -1 \neq 1 = \chi(h, k)\chi(g, h \oplus k)$. The associator functions $\phi = \partial F$ of the two algebras (defined by $(e_g e_h)e_k = \phi(g, h, k)e_g(e_h e_k)$) are both trivial on linearly dependent triples and are +1 on exactly 168 of the 420 unordered independent triples; neither is trilinear.*

Proof. The first claim is the definition: (2) and (1) differ only in $ac \rightarrow ca$, and $e_g e_h = -e_h e_g$ for distinct nonzero $g, h \in \mathbb{F}_2^3$. The cocycle computation is displayed. The counts and non-trilinearity are the finite computation of Appendix A; note that if $\phi_{\mathbb{S}}$ were trilinear it would equal $\det(v, \cdot, \cdot, \cdot)$ for some v , and the values $\phi_{\mathbb{S}}(1, 2, 8) = -1$, $\phi_{\mathbb{S}}(1, 2, 12) = \phi_{\mathbb{S}}(1, 4, 10) = \phi_{\mathbb{S}}(2, 4, 9) = 1$ force $v = 15$, contradicting $\phi_{\mathbb{S}}(9, 10, 12) = -1$. $\square$

Proposition 7.4. *The group of graded automorphisms of $\mathbb{S}'$ has order $2688 = 168 \cdot 16$; the permutation parts σ form the copy of $GL(3, 2)$ fixing V and ℓ , and every graded automorphism preserves V . The same holds for $\mathbb{S}$. There is no graded isomorphism $\mathbb{S} \rightarrow \mathbb{S}'$: for each of the 20160 elements $\sigma \in GL(4, 2)$ the linear system for λ is inconsistent.*

Proof. Computer search over $GL(4, 2)$ (Appendix A). The 16 sign changes $e_g \mapsto (-1)^{\mu(g)} e_g$, $\mu \in (\mathbb{F}_2^4)^*$, are automorphisms of every twisted group algebra of $\mathbb{F}_2^4$, and the $168 \cdot 8$ graded automorphisms of $\mathbb{O}$ [15] extend diagonally, so $2688 = 168 \cdot 16$ is the expected lower bound and the search shows it is attained. $\square$

Corollary 7.5. $\mathbb{S}$ and $\mathbb{S}'$ are not isomorphic, for each of the following independent reasons: (i) $\dim \mathcal{Z}$ is 11 versus 13; (ii) generic annihilators have dimension 4 versus 2; (iii) Aut is $G_2 \times S_3$ versus $G_2 \times \mathbb{Z}/2$; (iv) the alternative elements form a set with infinitely many maximal subspaces versus two.

8. THE MIRROR OCTONIONS

Everything above has a smaller model over $\mathbb{H}$, which is the quasi-octonion algebra of Cawagas and the algebra S_8 of Chan–Đoković.

Theorem 8.1. Let $M(\mathbb{H}) = \mathbb{H} + \mathbb{H}\ell$ with the product (2). Then:

- (1) $M(\mathbb{H})$ is unital, quadratic, flexible, not alternative, not a composition algebra, with $[x, x, y]$ given by (6); the spectrum of $L_{\bar{x}}L_x$ is $N \pm 2|a'| |b|$ (multiplicity 2 each) and N (multiplicity 4).
- (2) $x = a + b\ell$ is a zero divisor iff $a \in \text{Im } \mathbb{H}$ and $|a| = |b|$; $\mathcal{Z}(M(\mathbb{H})) = S^2 \times S^3$; every annihilator of a zero divisor is 2-dimensional, $\text{Ann}(u + b\ell) = \{nu + (bn)\ell : n \in \text{Im } \mathbb{H}, n \perp u\}$; and $\mathcal{P}(M(\mathbb{H})) \cong V_2(\mathbb{R}^3) \times S^3 \cong SO(3) \times S^3$ is a smooth 6-manifold.
- (3) $\text{Aut}(M(\mathbb{H})) \supseteq \{a + b\ell \mapsto \varphi(a) + (q\varphi(b))\ell : \varphi \in SO(3), q \in S^3\}$, a group of dimension $6 = \dim \text{Der}(M(\mathbb{H}))$; in particular the automorphism group acts transitively on $S^7 \cap \mathbb{H}\ell$, unlike in $\mathbb{S}'$, where $a + b\ell \mapsto a + (qb)\ell$ is an automorphism only for $q = \pm 1$.

Proof. (1) and (2) are the block computation on $M(Q)$ in the proof of Theorem 4.1 with $Q = \mathbb{H}$, together with the proof of Theorem 5.2, in which the associator condition is vacuous. (3): with $\mathbb{H}$ associative, $(a + (qb)\ell)(c + (qd)\ell) = (ca - \bar{d}qb) + (qda + qb\bar{c})\ell = (ca - \bar{d}b) + (q(da + b\bar{c}))\ell$. The dimension of Der is computed in Appendix A; that no $q \neq \pm 1$ works in $\mathbb{S}'$ is the nucleus argument of Theorem 7.1. $\square$

9. QUESTIONS

- (1) *Riemannian geometry.* $\mathcal{Z}(\mathbb{S}')$ is a product of round spheres; what is the induced metric on $\mathcal{P}(\mathbb{S}') \subset S^{15} \times S^{15}$ in the coordinates (u, n, q) ? It is G_2 -invariant of cohomogeneity three and not a Riemannian product (the cross terms between the S^3 and $V_2(\mathbb{R}^7)$ directions do not vanish). Is it Einstein for some rescaling, as Reggiani's deformations of $\mathcal{P}(\mathbb{S})$ are?
- (2) *The orientation tree.* Each word in $\{\text{CD}, \text{M}\}$ applied to $\mathbb{O}$ gives an algebra with the diagonal G_2 ; at dimension 16 the three algebras are $\mathbb{S}$, $\mathbb{S}'$ and $M(M(\mathbb{H}))$. Which of the 2^{n-3} words give non-isomorphic algebras at dimension 2^n , and which of them have $\text{Der} = \mathfrak{g}_2$? Numerical census in [16] over all sign functions on $\mathbb{F}_2^4$ with quaternionic lines indicates that $\mathbb{S}$ and $\mathbb{S}'$ are the only two with eight octaves and 14-dimensional derivation algebra; a proof would be desirable.
- (3) *Higher mirrors.* By Theorem 3.3, $M(\mathbb{S})$ and $M(\mathbb{S}')$ are subalgebras of the 64-dimensional Cayley–Dickson algebra. Do the results of [3] on large annihilators have mirror analogues, and is $\dim \text{Ann} \equiv 0 \pmod{2}$ rather than $\pmod{4}$ the general pattern?
- (4) *A conceptual proof of Theorem 5.5.* Moreno's proof that $\mathcal{P}(\mathbb{S}) \cong G_2$ realizes a zero-divisor pair as a Cayley frame. Is there a similarly intrinsic description of the trivialization $\mathcal{P}(\mathbb{S}') \cong V_2(\mathbb{R}^7) \times S^3$, e.g. via the fibration over $G_2/SO(4)$?

APPENDIX A. VERIFICATION

The computations referred to in the text were carried out in Python/NumPy/SymPy with the scripts accompanying this paper (`cd.py`, `f2iso.py`, `check1-check12`, and `sym.spectrum.py`). They construct $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}, \mathbb{S}, \mathbb{T}$ by iterated doubling (1), $\mathbb{S}'$ and $M(\mathbb{H})$ by (2), and the thirty-two Bales candidates by the corresponding formulas, and verify:

- unit, $x\bar{x} = N(x)$, flexibility, power-associativity up to degree 4, failure of alternativity and of the composition law, and the adjoint identity, for $\mathbb{S}$ and $\mathbb{S}'$;
- the annihilator dimensions of Theorem 5.2 on random and on structured points, the closed form of $\text{Ann}(u + b\ell)$, and the 2/6 stratification on a grid in the (b_0, b_1) disc;
- $\text{rank}(d\mu) = 16$ at points of every stratum of $\mathcal{P}(\mathbb{S}')$ ($\mu(x, y) = xy$), and tangent dimensions 14 for $\mathcal{P}$ and 13 for $\mathcal{Z}$ (respectively 14 and 11 for $\mathbb{S}$);
- the composition property of each of the fifteen hyperplane subalgebras of $\mathbb{S}$ and $\mathbb{S}'$ (Theorem 6.1) and of the thirty-one of $\mathbb{T}$;
- $\dim \text{Der} = 14$ for $\mathbb{S}, \mathbb{S}'$ and all thirty-two candidates, 6 for $M(\mathbb{H})$, the uniform P_1^T tower, and $M(M(\mathbb{H}))$;
- the symbolic characteristic polynomials of alt_x for $\mathbb{S}$ and $\mathbb{S}'$ in normal form, and the two block formulas (5), (6);
- the isomorphisms Φ of Theorem 3.3 for $A = \mathbb{H}, \mathbb{O}$, the six isomorphisms of Theorem 3.6, and the exhaustive graded-isomorphism search over $GL(4, 2)$ (Propositions 7.3, 7.4, Remark 3.8);
- the counts 84/112/336 of Proposition 5.4, Brown's order-three map on $\mathbb{S}$ and its failure on $\mathbb{S}'$, the automorphisms of Theorem 8.1(3), the failure of the quaternion-line condition for the twenty-four rejected formulas, the fact that $\ker \text{alt}_x$ is a composition subalgebra of $\mathbb{S}$, and the non-vanishing of the cross terms of the induced metric on $\mathcal{P}(\mathbb{S}')$ in the coordinates (u, n, q) (Section 9, Question 1).

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